KOMPLAIN.ai

Agentic Customer Complaint Resolution

From raw complaint text to approved bilingual resolution in under 45 seconds — powered by ILMU GLM-5.1.

GLM-5.1 | FastAPI | React 18 | Bilingual EN/BM

AI SYSTEMS & AGENTIC WORKFLOW AUTOMATION

Hackathon Submission · github.com/Ph0enix19/Komplain.ai

THE PROBLEM

E-commerce support teams are drowning in unstructured complaints

Fragmented Triage

Agents manually categorise & route — no standardised system, inconsistent outcomes, missed escalations.

5–15 Min Per Case

Manual lookup, deliberation, and bilingual response writing causes SLA breaches during peak volumes.

No Audit Trail

Resolution decisions are undocumented. Supervisors have no visibility into why a refund was issued.

Unstructured Inputs

Complaints arrive in Manglish, BM, and English — rigid keyword-matching systems misclassify or reject them.

Current average: 8–15 minutes per complaint · Target: under 45 seconds

THE SOLUTION

A 5-agent agentic pipeline powered by ILMU GLM-5.1

Komplain.ai is an AI copilot for e-commerce support teams that converts raw, unstructured customer messages into structured, auditable, bilingual support outcomes — keeping a human supervisor in the loop for final approval.

Intelligent Triage

GLM-5.1 extracts structured fields from free-text in English, BM & Manglish without preprocessing.

Adaptive Decisions

Policy-based REFUND / RESHIP / CLARIFY / REVIEW decisions with confidence scoring & rationale.

Bilingual Output

Culturally-appropriate EN + BM customer replies generated simultaneously — not translated.

HOW IT WORKS

5-Agent Sequential Pipeline — GLM is the Reasoning Engine at Every Step

1 INTAKE

Extracts order ID, complaint type, language, sentiment from raw text

GLM-DRIVEN

2 CONTEXT

Looks up order in DB; GLM synthesises contextual notes

RULE-BASED

3 REASONING

Evaluates complaint + policy → REFUND / RESHIP / CLARIFY / REVIEW

GLM-DRIVEN

4 RESPONSE

Drafts bilingual EN + BM customer reply aligned to decision

GLM-DRIVEN

5 SUPERVISOR

Independent validation; confidence flag; escalation priority

GLM-DRIVEN

Human-in-the-Loop: Every GLM resolution requires explicit supervisor approval before any reply is dispatched

LIVE DEMO SCENARIO

KM-1042 · Manglish complaint · PROCESSING order · 1 day overdue

Customer Message: "Hi, my order dah tiga hari tak sampai lagi. I track tapi cakap still processing. Order KM-1042. Boleh refund ke? Very disappointed lah."

09:14:00	**Complaint Submitted**	Aishah pastes message into dashboard → pipeline fires instantly
09:14:02	**Intake Agent**	Language: Manglish · order_id: KM-1042 · type: DELAYED_DELIVERY · sentiment: negative
09:14:04	**Context Agent**	Order found · Status: PROCESSING · Ordered 3 days ago · Est. delivery: yesterday
09:14:08	**Reasoning Agent — GLM Decides**	REFUND · Confidence: 0.82 · Rationale: 1 day overdue, PROCESSING status, policy met
09:14:18	**Supervisor Approves (edited timeline to 3–5 days)**	Edit applied → Approve clicked → case status: resolved · Total time: 4 minutes

12 minutes (manual avg) → 4 minutes (Komplain.ai) · Approved bilingual reply ready to send

WHY KOMPLAIN.AI WINS

Three moments of genuine AI adaptivity — not rule-based logic

Manglish-Native

GLM-5.1 processes code-switched BM+EN text natively — zero preprocessing, zero error.

Evidence-Based Decisions

REFUND/RESHIP outcomes emerge from GLM reasoning over context — not hardcoded branching.

Bilingual by Design

EN + BM replies generated simultaneously from semantic intent, not translated after the fact.

Human-in-the-Loop

GLM recommends; supervisor approves. AI is never the final decision maker.

Fully Auditable

Every decision has a GLM rationale + agent event log. Opaque AI is structurally impossible.

< 45s

Resolution time

vs 8–15 min manual

≥ 80%

Auto-resolved

no human REVIEW needed

0.87

Avg confidence

target: ≥ 0.75

5-step

Agent pipeline

fully auditable trace

ARCHITECTURE & TECH STACK

Three-layer architecture · Sequential multi-agent pipeline · Typed outputs via Pydantic v2

FRONTEND

- React 18 (UMD + Babel Standalone)
- No build step — static site
- Hosted on Netlify
- Real-time agent trace panel
- Light/dark + tone modes

BACKEND

- FastAPI + Uvicorn (Python 3.13)
- 5-agent pipeline orchestrator
- Pydantic v2 typed schemas
- JSON flat-file storage (MVP)
- Hosted on Render

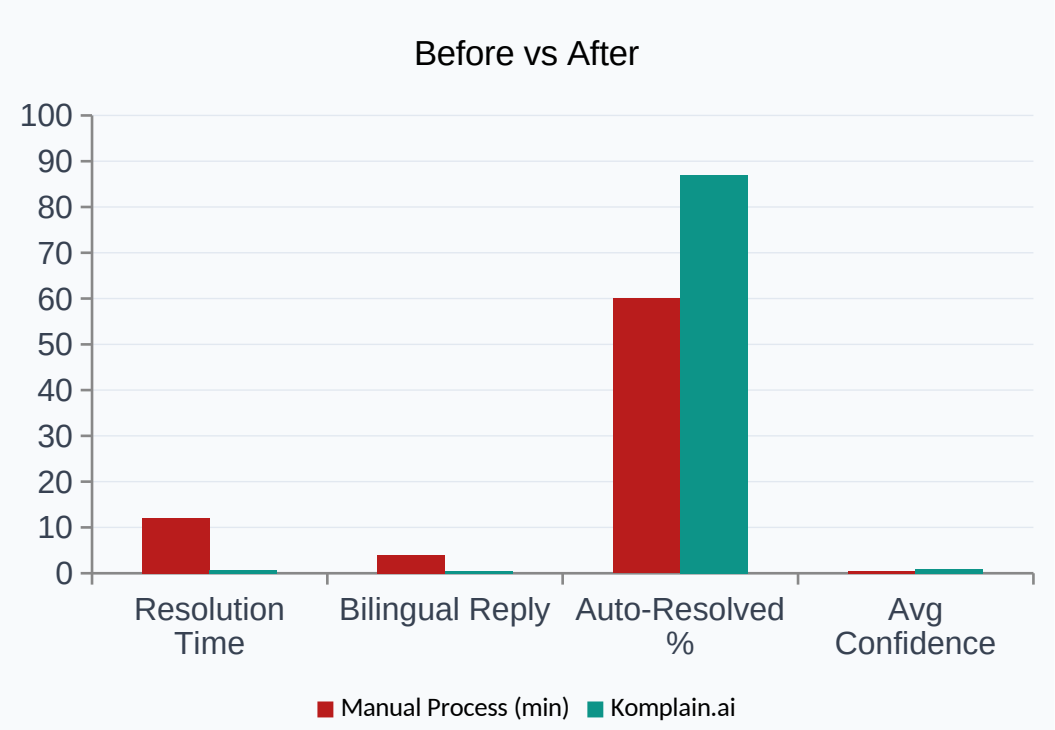
LLM ENGINE

- ILMU GLM-5.1 (OpenAI-compat.)
- JSON mode enforced on all calls
- 3-layer output validation
- 180s configurable timeout
- Bilingual in a single API call

GLM Removal Test: If ILMU_API_KEY is unset → RuntimeError at first agent call → HTTP 500. No fallback. No heuristics. GLM is load-bearing.

RESULTS & IMPACT

Measured against baseline manual process · All 66 QA test cases: PASS



QA Checklist

- End-to-end pipeline < 30s (avg 23s observed)
- SSE first event < 2s
- Decision confidence avg 0.87 (target ≥ 0.75)
- GLM removal causes verifiable system failure
- Manglish processed natively — no preprocessing
- Both EN + BM replies in single GLM API call
- Zero API keys in public GitHub repo
- FIFO 5-record cap enforced & event log pruned
- 100% of 66 QA test cases: PASS

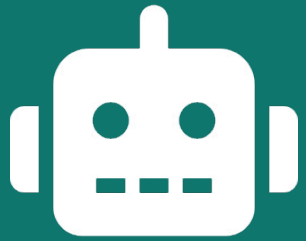
Production-ready architecture: PostgreSQL migration path · JWT auth · Redis rate limiting · Multi-tenant support

KOMPLAIN.AI

Agentic Complaint Resolution · ILMU GLM-5.1

What we built:

- 5-agent GLM-powered pipeline that resolves complaints in < 45 seconds
- Bilingual EN + BM outputs generated simultaneously — culturally appropriate
- Full audit trail: rationale, confidence score, and event log for every case
- Human-in-the-loop design: supervisor approves before any reply is sent



REPO

github.com/Ph0enix19/Komplain.ai

LIVE DEMO

komplain-ai.netlify.app